

Banking Document & Customer Support Assistant

Horizon National Bank · retrieval-augmented policy assistance with deterministic validation

01 PRESENTATION

Streamlit application

Home · Dashboard · Upload · Analysis · Policy AI · Sentiment · Voice · Search · Monitoring

MASKING BOUNDARY — account number, PAN, Aadhaar, customer ID and name are masked before this line. The model never receives an unmasked identifier.

02 IDENTITY

Microsoft Entra ID

DefaultAzureCredential → bearer token → every service. User-assigned managed identity `id-bankai-app` holds five data-plane roles.

Secrets in Key Vault (RBAC mode, purge protection). **No keys in code, no connection strings.**

03 PIPELINES

DOCUMENT PATH

Blob Storage / ADLS Gen2

raw-documents/{policies, loan, statements, kyc}

Text-layer probe → Document Intelligence

< 50 chars → prebuilt-read (OCR) · otherwise prebuilt-layout

Pydantic typed schema

"2 years" → 2.0 · "INR 75,00,000" → 7500000 · invalid = failure

RULE ENGINE — deterministic

Home Loan \$3 ≥ 3 yrs · \$4 ≥ INR 50,000 · \$7 ≥ 700 bureau

QUERY PATH

Question — typed or spoken

Speech-to-text, en-IN · aad#{resourceId}#{token} auth

Embedding — text-embedding-3-small

1536 dimensions · same deployment as the corpus

Azure AI Search — hybrid retrieval

104 chunks · BM25 + vector (HNSW) + semantic reranker

gpt-4.1-mini — explanation + citation

temperature 0 · strict JSON schema · refuses when uncovered

04 DECISION

VERDICT CROSS-CHECK

status ← rule engine, never the model. reason ← model explanation, which affects nothing.

Agreement raises confidence to HIGH. Disagreement lowers it to LOW and flags the case for manual review.

human_review_required = True — assigned in Python after generation. No code path sets it False.

05 AI SERVICES

One AIServices resource (S0, custom subdomain) fronts all five. Single endpoint, single managed-identity grant.

Document Intel.

layout · read
27 fields extracted

Azure OpenAI

gpt-4.1-mini
embedding-3-small

AI Search

Basic · HNSW
semantic reranker

AI Language

sentiment
PII detection

AI Speech

STT · TTS
en-IN

06 DATA

ADLS Gen2

raw-documents/ · processed/ · extracted-data/
Hierarchical namespace, TLS 1.2, no public access

Search index

bank-policies-idx · 104 chunks
Policies only. Customer records are never indexed.

Key Vault

RBAC authorisation mode
Purge protection, 7-day retention

07 GUARDRAILS

Four independent layers. An attack must defeat all four.

- | | |
|---------------|--|
| 1 PLATFORM | Azure AI Content Safety / Prompt Shields — returns HTTP 400 before the model sees the prompt |
| 2 INSTRUCTION | System prompt rules 1–8 — grounding, refusal string, retrieved text is data not commands |
| 3 STRUCTURAL | Strict JSON schema — the decision enum contains no APPROVED or REJECTED value |
| 4 CODE | Post-generation guards — prompt-leak suppression, human_review_required override |

Layer 4 exists because layer 2 proved unreliable: the assistant disclosed its system prompt on one run and refused on another, under identical settings.

The prompt states intent. Code enforces it.

08 TELEMETRY

OpenTelemetry spans → Application Insights → Log Analytics. Latency p50/p95, token spend, document success rate, grounding rate.

Spans emitted through the OTel API surface in the `dependencies` table, not `customEvents`. Automatic instrumentation captures every outbound Azure call.

MEASURED

RETRIEVAL — 10 questions, known ground truth

keyword top-1 7/10 recall@5 10/10 MRR 0.817

vector top-1 8/10 recall@5 8/10 MRR 0.800

hybrid top-1 8/10 recall@5 10/10 MRR 0.870

Expanding 7 → 12 policies reduced vector and improved keyword.
Hybrid absorbed both and held flat.

CORPUS

12 policy documents → 104 indexed chunks

20 customer applications, 11/11 fields each

6 image-only scans, zero text layer

60 customer messages, triaged

Synthetic, generated for internal consistency.

TESTS — 30 passing

10 RAG accuracy · 5 hallucination refusal

6 prompt injection, incl. indirect via corpus

5 sensitive data · masking · index PII

Structural: enum, code-enforced review flag

Caught a real prompt-disclosure bug.